

[scale=0.85]
[font=, black!80] at (16.0,23.5) Expansion Rotor Bank Angle & Pitch Depower Risk;
[shift=(0,3.0)] [black!1, rounded corners=6pt] (-0.5,-0.5) rectangle (15.0,19.0); [font=, black!70] at (7.5,19.0) ground station;
[shift=(5.5,12.0)] [thick] (-4.0,0) - (4.0,0); [font=] at (0,-0.5) ground station;
[gray!60, line width=1pt] (0,0) - (5.2,3.0); [font=, gray!60] at (6.0,3.3) shaft axis; [font=, gray!60] at (5.2,4.3) hub rotor (flat annulus);
[blue!60, line width=2pt] (4.3,2.5) - (6.1,3.5); [font=, blue!60] at (5.2,4.3) hub rotor (flat annulus);
[cyan!70, opacity=0.9] (2.2,1.27) circle (3pt);
[cyan!60!black, line width=2.0pt] (2.2,1.27) - (1.3,2.2); [cyan!60!black, line width=2.0pt] (2.2,1.27) - (1.3,2.2);
[cyan!40!black, line width=0.8pt, densely dashed] (1.3,2.2) arc (150:210:1.0);
[cyan!15, opacity=0.4] (2.2,1.27) - (1.3,2.2) arc (150:210:1.0) - cycle;
[-i, red!60, line width=1pt] (2.0,1.5) ++ (0.4,0.2) arc (40:85:0.8); [font=, red!60] at (1.2,2.0) bank = 45°;
[font=, cyan!70!black] at (0.5,0.7) expansion

rotor;
[-i, blue!50, line width=2.5pt] (-2.0,2.5) - (0.2,1.0); [font=, blue!50] at (-2.2,2.9) wind;
[green!40!black, line width=0.6pt, densely dotted] (0,0) - (2.2,1.27); [green!40!black, line width=0.6pt, densely dotted] (0,0) - (2.2,1.27);
[shift=(1.5,2.0)] [font=, black!60] at (2.0,2.5) Blade section (mid-span);
[black!70, line width=1.8pt, rotate around=-45:(0,0)] (0,0) to[out=10, in=180] (2.5,0.4) to[out=5, in=180] (2.5,0.4);
[-i, blue!60, line width=1.8pt] (-2.5,0.8) - (-0.1,0.0); [font=, blue!60] at (-2.5,1.2) apparent wind;
[-i, red!70!black, line width=2.0pt] (1.0,0.8) - (2.2,2.2); [font=, red!70!black] at (2.8,2.2) lift;
[-i, red!50, line width=1.5pt, dashed] (1.0,0.8) - (1.0,2.0); [font=, red!50!black] at (0.5,2.0) F_r (outward);
[-i, red!50, line width=1.5pt, dashed] (1.0,0.8) - (2.0,0.8); [font=, red!50!black] at (2.5,0.6) F_a (axial);
[-i, gray!50] (0.5,0.3) arc (130:175:0.7); [font=, gray!50] at (-0.1,0.9) 45°;
[font=, green!50!black, fill=green!5, draw=green!40!black, rounded corners=4pt, inner sep=4pt] at (1.5,1.27) expansion;
[font=, align=left, fill=black!3, draw=gray!30, rounded corners=5pt, inner sep=7pt] at (11.0,4.0) H

Blades are the cyan!70!blackouter portion of the rotor disk.
Unlike a flat-annulus main rotor, expansion blades
are red!60banked outward — they sweep a cyan!70!blackhollow truncated cone.
Lift vector resolves into:
• Radial F_r : spreads tethers outward
• Axial F_a : contributes to shaft compression
Net effect: reduces ring buckling → lighter shaft. ;

[shift=(18,3.0)] [red!2, rounded corners=6pt] (-0.5,-0.5) rectangle (15.0,19.0); [font=, red!50!black] at (7.5,19.0) ground station;
[shift=(5.5,12.0)] [thick] (-4.0,0) - (4.0,0); [font=] at (0,-0.5) ground station;
[gray!60, line width=1pt] (0,0) - (3.0,3.6); [font=, gray!60] at (3.8,3.8) shaft axis; [font=, gray!60, align=right] at (3.8,3.8) shaft axis;
(blackline out);
[blue!60, line width=2pt] (2.2,2.6) - (3.8,3.8); [font=, blue!60] at (3.4,4.6) hub rotor;
[red!70, opacity=0.9] (1.3,1.55) circle (3pt);
[red!60!black, line width=2.0pt] (1.3,1.55) - (0.3,2.5); [red!60!black, line width=2.0pt] (1.3,1.55) - (0.3,2.5);
[font=, red!70!black] at (-0.5,1.2) expansion

rotor;
[-i, red!50, line width=2.5pt] (-1.0,4.0) - (0.5,2.0); [font=, red!50] at (-1.5,4.3) wind hits

from above;
[font=, red!70, fill=red!3, draw=red!50, rounded corners=3pt, inner sep=3pt] at (4.0,1.0) ! BACK-WINDING;
[green!40!black, line width=0.6pt, densely dotted] (0,0) - (1.3,1.55); [green!40!black, line width=0.6pt, densely dotted] (0,0) - (1.3,1.55);
[shift=(1.5,2.0)] [font=, red!50!black] at (2.0,2.5) Blade section — back-winded;
[black!70, line width=1.8pt, rotate around=-45:(0,0)] (0,0) to[out=10, in=180] (2.5,0.4) to[out=5, in=180] (2.5,0.4);
[-i, red!60, line width=1.8pt] (1.5,2.5) - (0.2,0.2); [font=, red!60] at (1.8,2.8) apparent wind

(from above);
[-i, red!70!black, line width=2.0pt] (1.0,0.5) - (-0.2,-0.8); [font=, red!70!black] at (-1.2,-0.8) lift (REV);
[-i, red!50, line width=1.5pt, dashed] (1.0,0.5) - (0.7,-0.5); [font=, red!50!black] at (0.0,-0.5) F_r (inward);
[-i, red!50, line width=1.5pt, dashed] (1.0,0.5) - (1.8,0.8); [font=, red!50!black] at (2.5,0.8) F_a ;
[font=, red!70!black, fill=red!5, draw=red!50!black, rounded corners=4pt, inner sep=4pt] at (1.5,-1.27) expansion;
[font=, align=left, fill=red!3, draw=red!40, rounded corners=5pt, inner sep=7pt] at (11.0,4.0) Why

Pitch depower tilts the shaft upward (50°).
Wind direction is unchanged — but the blade
orientation relative to wind flips.
At 45° bank, apparent wind hits from above.
Lift reverses: radial force pulls tethers INWARD.
 r_{eff} drops $\propto r^2$ — torsional capacity collapses.
Result: shaft can buckle during shutdown. ;

[red!8, draw=red!50!black, line width=1.5pt, rounded corners=5pt] (-0.5,0.0) rectangle (33.0,2.8); [font=, black!70] at (16.0,-0.5) Mitigation options;; [font=, align=center, black!60] at (16.0,-1.1) Low